

Dionne Argyropoulos

PHD

Infecton and Global Health Division, Walter & Eliza Hall Institute

she/her | +61 411 394 914 | argyropoulos.d@wehi.edu.au | 0000-0002-8068-0215 | [cISfd8kAAAAJ](#) | [dionnecargy](#) | [dionnecargy](#)

About Me

I am a molecular epidemiologist and population geneticist currently positioned at WEHI (Walter and Eliza Hall Institute). I have an avid interest in genetics, evolution and coding which led me to apply these approaches to studies of infectious diseases, such as *Plasmodium falciparum* malaria in my Masters of Science and PhD, to *P. vivax* malaria and various microbes in my current first post-doc position. Research topics of interest include drug or antimicrobial resistance, spatiotemporal trends, epidemiological statistics and machine learning techniques, data wrangling and visualisation. My work has involved international collaborations in East and West Africa, North and South America and Southeast Asia, as well as supervision and support of students, with various opportunities for knowledge exchange and capacity-building in coding and genetic epidemiology. Being an active member of the R-Ladies+ Melbourne community, and the current President (2024-2026), has led to the development of practical content (workshops and the website) and has motivated me to learn and develop new techniques. Throughout my research projects, I have developed a deep interest in coding and have even generated this CV using R/Quarto!

Current Position

Walter and Eliza Hall Institute of Medical Research (WEHI)

Melbourne, AU

RESEARCH OFFICER

2024-current

- Post-doctoral researcher with a focus on machine learning and advanced epidemiological methods.

Education

University of Melbourne

Melbourne, AU

DOCTOR OF PHILOSOPHY - MEDICINE, DENTISTRY AND HEALTH SCIENCES

2020-24

- Recipient of the Research Training Program Scholarship (Stipend and Fee Offset, 2020-2024) awarded to high-achieving domestic research students with a full fee offset and living allowance.
- My thesis, "The genetic epidemiology of the *Plasmodium falciparum* malaria in high burden settings in Africa", focussed primarily on molecular epidemiology and population genetics.
- I compared existing and developed new methods optimised for low-density and multiclonal datasets found in asymptomatic infections in high-transmission settings in Africa.
- I analysed the effect of various malaria control interventions on the population genetics of asymptomatic *Plasmodium falciparum* infections in an area of high seasonal malaria transmission in Ghana.

University of Melbourne

Melbourne, AU

MASTER OF SCIENCE (BIOSCIENCES)

2018-19

- Coursework and Research Project.
- My thesis was entitled "Understanding the effects of perturbations on the population genetics of asymptomatic *Plasmodium falciparum* infections in an area of high seasonal malaria transmission in Ghana".
- I identified minimal variation in putatively neutral microsatellite genetic diversity between seasons and following a vector control intervention.
- Graduated with Distinctions and Honours in First Class

University of Melbourne

Melbourne, AU

BACHELOR OF SCIENCE

2015-17

- Major in Genetics, with supplementary courses in Biochemistry and Molecular Biology, Evolution and Ecology, Reproductive Biology and Demographics.
- Graduated with Honours in First, Second and Third Class.

Publications

Longitudinal analysis of the prevalence of minor *Plasmodium* spp. in the reservoir of asymptomatic infections through sequential interventions in Northern Sahelian Ghana

PLoS Neglected Tropical Diseases

RIOS-TERAN, C. A., TIEDJE, K. E., BANGRE, O., DEED, S. L., ARGYROPOULOS, D. C., KORAM, K. A., ODURO, A. R., ANSAH, P. O. &

DAY, K. P DOI.ORG/1371/JOURNAL.PNTD.0014174

2026

Navigating parasite antigen genetic diversity in the design of *Plasmodium vivax* serological exposure markers for malaria

bioRxiv

BARENG, P., WU, K. W., TAKASHIMA, E., ARGYROPOULOS, D., SMITH, L., NAUNG, M., MAZHARI, R., SCHOFFER, K., KIERNAN-WALKER, N., ABRAHAM, A., LAMONT, M., MEHRA, S., LIM, P., SATTABONGKOT, J., MONTEIRO, W., LACERDA, M., HEALER, J., CHITNIS, C. E., THAM, W.-H., TSUBOI, T., MUELLER, I., BARRY, A. E. & LONGLEY, R. J.

2025

DOI.ORG/1101/2025.08.15.25333797

Evaluating the Performance of Predicting *Plasmodium vivax* Infection Risk Using Serological Markers in Patients with *Plasmodium falciparum* Malaria

Journal of Infectious Diseases

HAFIDZAH, M., DEGAGA, T. S., CHRISTIAN, M., ALAM, M. S., HOSSAIN, M. S., BAIRD, J. K., SUTANO, I., SMITH, L., ARGYROPOULOS, D., AMELIA, A. R., NOVIYANTI, R., PRICE, R. N., MUELLER, I., THRIEMER, K. & LONGLEY, R. J.

2025

DOI.ORG/1093/INFDIS/JIAG164

Measuring changes in *Plasmodium falciparum* census population size in response to sequential malaria control interventions

eLife

TIEDJE, K. E., ZHAN, Q., RUYBAL-PÉSANTÉZ, S., TONKIN-HILL, G., HE, Q., TAN, M. H., ARGYROPOULOS, D. C., DEED, S. L., GHANSAH, A., BANGRE, O., ODURO, A. R., KORAM, K. A., PASCUAL, M. & DAY, K. P. DOI.ORG/7554/ELIFE.91411.4

2025

Performance of SNP barcodes to determine genetic diversity and population structure of *Plasmodium falciparum* in Africa

Frontiers in Genetics

ARGYROPOULOS, D. C., TAN, M. H., ADOBOR, C., MENSAH, B., LABBÉ, F., TIEDJE, K. E., KORAM, K. A., GHANSAH, A. & DAY, K. P.

2023

DOI.ORG/3389/FGENE.2023.1071896

Comparison of molecular surveillance methods to assess changes in the population genetics of *Plasmodium falciparum* in high transmission

Frontiers in Parasitology

GHANSAH, A., TIEDJE, K. E., ARGYROPOULOS, D. C., ONWONA, C. O., DEED, S. L., LABBÉ, F., ODURO, A. R., KORAM, K. A., PASCUAL, M. & DAY, K. P. DOI.ORG/3389/FPARA.2023.1067966

2023

Neutral vs. non-neutral genetic footprints of *Plasmodium falciparum* multiclonal infections

PLoS Computational Biology

LABBÉ, F., HE, Q., ZHAN, Q., TIEDJE, K. E., ARGYROPOULOS, D. C., TAN, M. H., GHANSAH, A., DAY, K. P. & PASCUAL, M.

2023

DOI.ORG/1371/JOURNAL.PCBI.1010816

Indoor residual spraying with a non-pyrethroid insecticide reduces the reservoir of *Plasmodium falciparum* in a high-transmission area in northern Ghana

PLoS Global Public Health

TIEDJE, K.E., ODURO, A.R., BANGRE, O., AMENGA-ETEGO, L., DADZIE, S.K., APPAWU, M.A., FREMPONG, K., ASOALA, V., RUYBAL-PÉSANTÉZ, S., NARH, C.A., DEED, S.L., ARGYROPOULOS, D. C., GHANSAH, A., AGYEI, S.A., SEGBAYA, S., DESEWU, K., WILLIAMS, I., SIMPSON, J.A., MALM, K., PASCUAL, M., KORAM, K.A. & DAY, K.P. DOI.ORG/1371/JOURNAL.PGPH.0000285

2022

The impact of indoor residual spraying on *Plasmodium falciparum* microsatellite variation in an area of high seasonal malaria transmission in Ghana, West Africa

Molecular Ecology

ARGYROPOULOS, D. C., RUYBAL-PESÁNTÉZ, S., DEED, S.L., ODURO, A.R., DADZIE, S.K., APPAWU, M.A., ASOALA, V., PASCUAL, M., KORAM, K.A., DAY, K.P. & TIEDJE, K.E. DOI.ORG/1111/MEC.16029

2021

Teaching and Supervision

Alongside the positions outlined below, I have been actively involved in the fostering collaboration with researcher officers from the Noguchi Memorial Institute for Medical Research in Ghana (2021-2023) to develop skills in R, population genetics, PCR genotyping and epidemiology. Furthermore, I have worked with the research officers at the University of Chicago, USA (2021-2023) to further expand skills in infectious disease modelling and theoretical population genetics and ecology.

Trained IPC Team in R Shiny and R Package for Serology Data Analysis

WEHI, INSTITUT PASTEUR CAMBODIA

2026

- Knowledge Transfer of Serology Data Analysis in R with team in IPC.

Workshop Developer, Tutor, Demonstrator for BIOL90042

WEHI, UoM

2026

- Aided development of machine learning workshop and proof-read other workshop materials for R-based coding subject. Taught machine learning workshop.

Tutor and Demonstrator for Introduction to R Course, WEHI

WEHI

2026

- Delivered ggplot2 plotting tutorial and assisted in tutoring WEHI students and staff in R coding.

Co-supervisor of PhD Student Alexandra Steward

WEHI

2025

- Co-supervisor with experience in molecular biology (PCR) lab work, bioinformatics and genetics data analyses

Tutor for Introduction to R Course, WEHI

WEHI

2025

- Assisted in tutoring students and staff in R coding.

Supervised International Collaborator and Student from Ethiopia

WEHI

2025

- Managed progress, trained in PvSeroApp and data analysis, invited to meetings (19 Mar - 15 May 2025)

Co-supervisor of undergraduate pathology research subject

DEPARTMENT OF IMMUNOLOGY AND MICROBIOLOGY, UoM

2023

- Assisted in training an undergraduate student in genomic epidemiology and population genetics, performing PCR and using R to generate statistics.

Co-supervisor of undergraduate pathology research subject

DEPARTMENT OF IMMUNOLOGY AND MICROBIOLOGY, UoM

2022

- Assisted in training an undergraduate student in genomic epidemiology, population genetics and bioinformatics, focussing on vector competence genes, developing a novel PCR and data mining from an online repository (MalariaGEN Pf6: [↗](#))

First Year Biology Assistant/Supervising Demonstrator and Tutor

FACULTY OF SCIENCE, UoM

2018-2021

- Laboratory assistant demonstrator, teacher and mentor for first-year Biology students at the UoM.

First and Second Year Genetics Marker

FACULTY OF SCIENCE, UoM

2020-2021

- Providing feedback for students and marking assessment tasks relating to various Genetics courses.

First Year Biology Marker

FACULTY OF SCIENCE, UoM

2021

- Providing feedback for students and marking assessment tasks relating to various First Year Biology courses.

Evolution and the Human Condition Marker

FACULTY OF SCIENCE, UoM

2021

- Providing feedback for students and marking assessment tasks.

VCE Biology Demonstrator

FACULTY OF SCIENCE, UoM

2018-2019

- Laboratory demonstrating and teaching for Year 12 VCE Biology Students.

VCE Biology Issues in Gene Technology Workshop

FACULTY OF SCIENCE, UoM

2019

- I ran a 1.5-hour workshop teaching Year 12 Biology students about Gene Technology in an agricultural context.

ITAS Tutor

MURRUP BARAK, UoM

2019

- One-on-one tutoring for university students in the Melbourne Institute of Indigenous Development.

Conference Presentations

- 2026 **posit::conf(2026)** (Scholarship): Conference to discuss latest developments in R and Python. [↗](#).
- 2025 **European Summer Program in Infectious Disease Analysis and Modelling (ESPIDAM)** (Poster): Summer program held for PhD students, post docs and public health scientists focussed on infectious disease modeling and epidemiology. Presented poster on machine learning algorithm implemented in PvSeroTAT.
- 2025 **Malaria Seminar** (Speaker): Organised by Malaria in Melbourne committee, with on bimonthly talks on latest malaria research. Presented 20 minute talk on PvSeroTAT work.
- 2024 **Genomic Epidemiology of Malaria** (Speaker/Poster): International forum for malaria scientists, clinicians, and public health professionals working at the interface of genome science and technology, epidemiology and statistical and population genetics [↗](#).
- 2024 **Molecular Approaches to Malaria Conference** (Poster): International conference on latest developments in malaria research [↗](#).
- 2023 **Malaria in Melbourne Conference** (Poster): Australian biennial conference of all aspects of malaria research [↗](#).
- 2023 **International Congress of Genetics** (Poster): The international genetics community came together to share ideas and the latest research under the overarching theme: Genetics and Genomics: Linking Life and Society [↗](#).
- 2022 **Peter Doherty Institute/POSSIIM Graduate Research Conference** (Speaker): Conference for all students in the department to showcase work. 10-minute oral presentation.
- 2022 **MDHS Graduate Research Conference** (Speaker): Conference on latest developments in MDHS faculty at UoM. 15-minute oral presentation to faculty.
- 2021 **DMI Graduate Research Symposium** (Speaker): Conference for all students in DMI-Doherty PhD program. I presented a five-minute talk on PhD outcomes.
- 2021 **Malaria in Melbourne Conference** (Speaker/Poster): Australian biennial conference of all aspects of malaria research. Presented three-minute short talk and poster [↗](#).
- 2021 **Victorian Infection and Immunity Network Young Investigator Symposium** (Speaker): VIIN conference for post-graduate students, post-doctorate fellows and research assistants. 10-minute talk.
- 2020 **Molecular Approaches to Malaria Conference** (Poster): International conference on latest developments in malaria research [↗](#).
- 2019 **Malaria in Melbourne Conference** (Speaker/Poster): Australian biennial conference of all aspects of malaria research. Offered for 15-minute talk, presented poster on MSc research. [↗](#).
- 2019 **Victorian Infection and Immunity Network Young Investigator Symposium** (Poster): VIIN conference for post-graduate students, post-doctorate fellows and research assistants. Presented a poster on MSc research [↗](#).
- 2019 **Stockholm International Youth Science Seminar Panel Speaker** (Panel/Poster): One of three students in one-hour panel on the topic "The Scientific Process" with moderator Anna Wetterbom, the CEO of the Young Academy of Sweden. Panel was broadcast online to an international audience (Facebook, YouTube). Presented poster on MSc research to high school students.
- 2019 **Victorian Biodiversity Conference** (Poster): Annual scientific conference focussed on highlighting biodiversity-related research and management projects based in Victoria, AU. Presented poster on MSc research [↗](#).
- 2018 **BioSciences Post-Graduate Symposium** (Speaker): Students and staff of the School of BioSciences showcased their research undertaken.

Skills

Data Analysis: Population Genetics, Epidemiology, Machine Learning

Laboratory: PCR Development/PCR, DBS cutting, DNA Extraction, Gel Extraction

Programming: Advanced R, Quarto, Rmarkdown, Git/GitHub, Beginner (LaTeX, Python, Bash), Intermediate (Linux/Command-Line)

Soft-Skills: Science communication, Coaching and Mentoring, Team member, Organised

Software: Mac OSX Platform, Microsoft Office, BOTTLENECK, GeneMarker

Web: GitHub Pages, BLAST, R Shiny

Digital tools

I have also developed a complex R Shiny web application to support *Plasmodium vivax* surveillance efforts.

PvSeroApp

D ARGYROPOULOS

2025

- The PvSeroApp Shiny web application streamlines the data processing of the multi-antigen Luminex-based Plasmodium vivax serological data and applies the machine learning classification algorithm to identify individuals with recent exposure to P. vivax. [↗](#)

SeroTrackR

D ARGYROPOULOS

2025

- SeroTrackR R package provides the backbone for the PvSeroApp Shiny web application, with data processing of the multi-antigen serological data and applies the machine learning classification algorithm to identify individuals with recent exposure to P. vivax. [↗](#)

wastewaterR

D ARGYROPOULOS

2025

- wastewaterR R package contains helper functions relevant for wastewater-specific qPCR projects for in-house and collaborator needs. [↗](#)

melbourne

D ARGYROPOULOS

2025

- Developed melbourne R package with colour palette of melbourne tram network. [↗](#)

Awards and Funding

- 2026 **Opportunity Scholar for posit::conf(2026)** (Posit PBC): Awarded full scholarship to attend the posit::conf(2026) in Houston, USA inclusive of travel, incidentals, hotel, conference and workshop registration [↗](#).
- 2025 **Women in Malaria 2025 Prize for Free Registration in 2026** (Women in Malaria (WiM) Conference): Won a raffle to have free attendance for the 2026 conference.
- 2025 **WEHI Postdoctoral Association Professional Development Award** (WEHI Postdoctoral Association (PDA)): Awarded A\$1,500 to attend the European Summer Program in Infectious Disease Analysis and Modelling (ESPIDAM).
- 2025 **ACREME MASTER-MAP Travel and Training Award** (Australian Centre of Research Excellence in Malaria Elimination (ACREME)): Awarded A\$2,00 to attend the European Summer Program in Infectious Disease Analysis and Modelling (ESPIDAM). [↗](#) "Write-up here"
- 2024 **Graduate Researcher Conference Support Program** (The University of Melbourne): Awarded A\$1,500 to attend the Genetic Epidemiology of Malaria conference.
- 2024 **Major Bartlett University Scholarship** (The University of Melbourne): Awarded A\$2,000 to attend the Genetic Epidemiology of Malaria conference [↗](#)
- 2021 **Malaria in Melbourne (MiM) Short Talk Prize** (Malaria in Melbourne (MiM) Conference): Selected to attend Infection and Immunology conference in Lorne 2022.
- 2019 **In2Science Invitation to Speak about Experience** (In2Science): The only invited mentor to discuss my experience in the In2Science program over various years and engaging with students in STEM topics. [↗](#)
- 2019 **In2Science Nomination for Impact Mentor Award** (In2Science): Nominated as one of the five mentors in Victoria to have demonstrated a significant positive impact on engaging with students in STEM. [↗](#)
- 2019 **Leaders in Communities Award (LiCA)** (The University of Melbourne): Awarded after completing over 20 hours of volunteer work during my Master of Science degree. [↗](#)
- 2019 **Student Ambassador Leadership Program** (The University of Melbourne): Selected as a Student Ambassador for the Faculty of Science. [↗](#)
- 2019 **Stockholm International Youth Science Seminar Representative** (The University of Melbourne): Selected as the only Australian student to represent the UoM. 25 countries selected one student to attend Nobel Prize Week events (including Ceremony and Banquet) and speak about their research. [↗](#)
- 2019 **Communication for Research Scientists Three Minute Thesis (3MT) Runner-Up** (The University of Melbourne): Entire cohort for this subject (150) presented a 3MT on their research, with the intention of communicating to a general science audience.
- 2018 **BioSciences Post-Graduate Symposium 10-minute talk Runner Up** (The University of Melbourne): All PhD students, Masters, Honours students and staff of the School of BioSciences shared their research. I came second of 45 participants.
- 2018 **In2Science Role Model Mentor Award Finalist** (In2Science): Of 276 mentors around Victoria in 2018, 4 were finalists: one mentor per university in the program. I was the UoM finalist.

WEHI Postdoctoral Association (PDA)	Communications Officer, Board Member
WEHI POSTDOCTORAL ASSOCIATION (PDA) Board member attending monthly 1-hour meetings discussing and advocating postdoctoral issues and events. Communications role focusses on generating a topical monthly newsletter showcasing grants, travel awardees, events, and other articles.	2026
Malaria in Melbourne (MiM)	Chair
MALARIA IN MELBOURNE (MiM) Chaired first plenary session (Dr Steven Kho) and Short Talks 1 of the MiM Conference (30 Oct 2025).	2025
Australian Centre of Research Excellence in Malaria Elimination (ACREME)	Meeting Co-organiser
AUSTRALIAN CENTRE OF RESEARCH EXCELLENCE IN MALARIA ELIMINATION (ACREME) Multiple 1-hour meetings throughout the year to help organise the two-day meeting from 06 Feb to 14 Oct 2025.	2025
Australian Centre of Research Excellence in Malaria Elimination (ACREME)	Chair
AUSTRALIAN CENTRE OF RESEARCH EXCELLENCE IN MALARIA ELIMINATION (ACREME) Chaired final session "Updates from ACREME Investigators: Major projects" on 14 Oct 2025.	2025
World Mosquito Day	Poster Marker
WORLD MOSQUITO DAY Assessed scientific posters at the World Mosquito Day symposium 20 Aug 2025.	2025
R-Ladies+ Melbourne	Speaker
R-LADIES+ MELBOURNE Presented about R-Ladies+ Melbourne's involvement in education and teaching, especially in the It Takes a Spark STEM conference on 26 Feb 2025. 🔗 "MeetUp Event"	2025
International Day of Women in Statistics and Data Science	Panelist
INTERNATIONAL DAY OF WOMEN IN STATISTICS AND DATA SCIENCE 1-hour panel to discuss the work R-Ladies+ Melbourne and R-Ladies+ focus on and advocate for: 🔗 "View Talk on YouTube"	2025
R Consortium: R Dev Day	Participant
R CONSORTIUM Full-day experience to work on issues relating to R with the R Consortium. First Australian R Dev Day. Worked on dashboard development. 20 Nov 2025.	2025
Women in STEM: Inspiring the Next Generation of Girls in STEM	Mentor
VESKI Mentor for two students at a high school; I supervised and taught them about malaria and public health. They wrote a project about this work. 6x1-hour sessions with students, organised a half-day visit to WEHI and attended final ceremony with students.	2025
R-Ladies Melbourne	President
R-LADIES MELBOURNE INC Key administrator and leader in organising events and liaising with speakers, committee and group. Organised committee meetings and events, executive decision maker, liaised with collaborators and other non-for-profit organisations, liaised with funding bodies (DiUS, Posit).	2024-2026
It Takes A Spark STEM	Speaker
R-LADIES MELBOURNE INC Co-Speaker and Co-creator of Material for 80-minute Problem Solver Session 🔗 "Become A Disease Detective: Solve An Outbreak" 🔗	2023-2025
Careers in Data Science and Statistics Panel	Speaker
R-LADIES MELBOURNE INC Moderator of Panel discussing careers in data science and statistics, focussing on R and coding languages.	2024
R-Ladies Melbourne	Vice President
R-LADIES MELBOURNE INC Administrator and assistant to president to organise events and liaising with speakers, committee and group.	2023-2024
R-Ladies Melbourne	Committee Member
R-LADIES MELBOURNE INC Frequent attendee of all meetings and events.	2023